

A Fixed-Past Theorem for Count-Triggered Draw Rules in Finite Games, with Exhaustive Verification

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Abstract

In games such as chess, a position that has occurred before may be claimed as a draw, and a position that has occurred often enough is drawn automatically. The value of a position therefore depends on the history that reached it. We prove that this dependence is bounded. For a finite deterministic two-player reachability game with neutral, count-triggered draw rules, a monotone move clock and exactly valued irreversible exits, the win/draw/loss value of a query with an arbitrary supplied past is determined by two frozen threshold sets of that past, with no updating of occurrence counts during the solve. The proof identifies the natural “latest winning clock” function with the attractor rank of a reachability game on the position keys, from which a winning strategy that is positional in the key alone follows; that strategy never revisits a key, so no future repetition event can arise, including a defender’s option to claim before making a repeating move. We show by machine-checked counterexamples that each hypothesis is necessary; rewarded repetition, penalised repetition, a non-monotone clock, and non-alternating ownership (for the one-mask specialisation) each break the equivalence. An independent clean-room implementation of the explicit-history game and of the reduced game agrees on every one of more than 8×10^8 queries, including complete enumeration of all two-key and three-key arenas in a defined family. This note is the abstract companion to a manuscript instantiating the result for FIDE chess.

1 Introduction

A transposition table identifies different plays that reach the same position. Repetition rules make those plays differ in value; the same board with the same side to move may be drawn if it has occurred twice before and winning otherwise. This is the graph-history interaction problem of Campbell [1] and Kishimoto and Müller [2, 3]. Engine practice handles it heuristically (Stockfish scores a third occurrence relative to game history as a draw), and tablebase authors have observed informally that a metric-optimal winning line never repeats a position, so that only history supplied from outside the search can spoil a tablebase value [4].

This note asks a narrow question and answers it with a theorem rather than a heuristic. Fix a query, that is, a position, a move-clock value, and an arbitrary past given as a count of earlier occurrences for every position. Must an exact solver from that query keep updating occurrence counts along every hypothetical continuation? It need not. Two sets of positions suffice: those the past has already brought to within one occurrence of a claimable draw, and those within one occurrence of an automatic draw. Both are frozen for the whole solve. Under strict alternation of ownership, one set suffices once the query position’s own automatic status is checked.

The mechanism is elementary once named. The reduced game is a reachability game with a step budget; its “latest winning clock” is an affine image of the classical attractor rank [5, 6]; an attractor strategy strictly decreases rank and therefore never revisits a key. The only step beyond folklore is that the ranking property at *defending* keys holds for *every* successor, which excludes a defender’s option to claim a draw before executing a move that would repeat a key, a rule that chess has (Article 9.2.1) and that engine heuristics do not model.

Contributions. (i) A precise statement of the fixed-past equivalence for a general class of count-triggered neutral spoilers (Theorem 10), with the two-threshold chess-style rules and the one-mask alternating case as corollaries. (ii) The identification of the deadline function with the attractor rank and the resulting key-positional winning strategy (Proposition 6, Corollary 8). (iii) Machine-checked counterexamples showing that neutrality, clock monotonicity and (for the one-mask case) alternation are each necessary (Section 5). (iv) Exhaustive verification over all arenas of a defined two- and three-key family and over dense random arenas, with zero disagreements in more than 8×10^8 queries (Section 6). We claim no priority over the graph-history literature; what is new here is the exact combined statement and its verification, not the ingredients.

2 Model

Definition 1 (Epoch arena). An epoch arena is a finite set Q of nonterminal *keys*, a finite set T of terminals each labelled with an outcome in $\{0, 1, D\}$ (a win for player 0, a win for player 1, or a draw), an owner $o(q) \in \{0, 1\}$ for each key, and for each key two action sets: *quiet* moves to successors in $Q \cup T$, and *exits*, each labelled with an exact outcome in $\{0, 1, D\}$. A key with no action is a draw.

Keys are exact, meaning owner, actions and terminal status are functions of the key, and the key is the unit that repetition counts. Exits abstract irreversible moves (captures and pawn moves in chess) into already-solved subgames; their outcome is assumed independent of the history and clock at the moment of exit. Composition across exits is a separate, standard rank argument and is not treated here.

Definition 2 (Rules). Fix clock thresholds $1 \leq C < H$ and, for every key x , repetition thresholds $2 \leq r_x < a_x$. A *query* is (q_0, h_0, c) with $q_0 \in Q$, $0 \leq h_0 \leq H$, and a past count $c : Q \rightarrow \mathbb{N}$ excluding the current occurrence of q_0 .

Definition 3 (Explicit-history game). States are (q, h, N) with $q \in Q \cup T$, $0 \leq h \leq H$ and $N : Q \rightarrow \mathbb{N}$ the occurrence counts *including* the current occurrence. The initial state is $(q_0, h_0, c + \mathbf{1}_{q_0})$. At a terminal the game ends with its outcome. At a nonterminal (q, h, N) : if $h = H$ or $N(q) \geq a_q$ the game ends in a draw; otherwise the owner $o(q)$ chooses among (a) a quiet move to q' , leading to $(q', h + 1, N + \mathbf{1}_{q'})$; (b) an exit, ending with its outcome; (c) a *claim*, ending in a draw, available iff $N(q) \geq r_q$, or some quiet successor q' satisfies $N(q') + 1 \geq r_{q'}$, or $h \geq C$, or some quiet move exists and $h + 1 \geq C$. Claims are optional actions, not compulsory outcomes.

Because h strictly increases along quiet moves and every other action terminates, the explicit game is a finite acyclic game and is determined by backward induction. Counts may be capped at a_x without changing values.

Definition 4 (Frozen-mask game). Given a query, let $B = \{x : c(x) \geq r_x - 1\}$ and $A = \{x : c(x) \geq a_x - 1\}$, so $A \subseteq B$. The frozen-mask game has states (q, h) ; at a nonterminal, $h = H$ or $q \in A$ is a draw; a claim is available iff $q \in B$, or some quiet successor lies in B , or $h \geq C$, or a quiet move exists and $h + 1 \geq C$. Quiet moves lead to $(q', h + 1)$. A and B never change.

The frozen-mask claim set is included in the explicit claim set at the initial occurrence of each key (an explicit self-loop intended claim with $c(q) = r_q - 2$ is omitted; it never arises on a winning line, see Lemma 7). The frozen-mask game never invents a claim that the explicit game lacks.

3 Deadlines are attractor ranks

Fix a target player P . P wins a state if P can force a P -won terminal or exit; draws and wins for the other player both refute this objective, so P never claims and the opponent may.

Lemma 5 (Clock monotonicity). *In the frozen-mask game, if P wins (q, h) then P wins (q, h') for every $h' \leq h$.*

Proof. Couple plays from (q, h') and (q, h) move for move. Quiet moves keep the clocks a constant offset apart; repetition claims depend only on the frozen masks and coincide; clock claims available at the lower clock are available at the higher; exits and terminals coincide. Every opponent action in the lower-clock play is therefore available in the higher-clock play, so the winning strategy from (q, h) transfers. $\square$

Define the *deadline* $D_P(q)$ as the largest h with (q, h) winning for P , or -1 if none; $D_P = H$ at P -won terminals and -1 at other terminals. Let $\text{sat}(z) = \max(-1, z)$, and for a candidate vector d let the action value be $w_d(e) = \text{sat}(d(q') - 1)$ for a quiet move to q' , $H - 1$ for a P -won exit and -1 otherwise. Let $K_P(q) = H - 1$ at P -owned keys and $K_P(q) = \min\{H - 1, C - 1 - \mathbf{1}[q \text{ has a quiet move}]\}$ at opponent keys (the last clock at which no clock claim is available). Let Z_P be the set of keys in A , actionless keys, and opponent keys at which a repetition claim of Definition 4 is available. The operator

$$(F_P d)(q) = \begin{cases} -1 & q \in Z_P, \\ \min\{K_P(q), \max_e w_d(e)\} & o(q) = P, \\ \min\{K_P(q), \min_e w_d(e)\} & o(q) \neq P, \end{cases} \quad (1)$$

is monotone on the finite lattice $\{-1, \dots, H - 1\}^Q$ and D_P is its unique fixed point with the prescribed terminal values (uniqueness by descending induction on the clock; each integer coordinate is determined by its Boolean threshold layers).

Proposition 6 (Affine identification). *Let $\rho(q) = H - 1 - D_P(q)$, with $\rho = H$ when $D_P = -1$. Then ρ is the min-max distance to a P -winning target on the key arena with unit step cost, node-dependent lower bounds $H - 1 - K_P(q)$ at opponent keys, and target set the P -won exits and terminals: at P -keys $\rho(q) = \min_e(\rho(q') + 1)$ and at opponent keys $\rho(q) = \max(H - 1 - K_P(q), \max_e(\rho(q') + 1))$, saturating at H . In particular ρ is the attractor rank of a reachability game [5], and (q, h) is winning iff $h + \rho(q) \leq H - 1$.*

Proof. Substitute $d = H - 1 - \rho$ in (1); $\text{sat}(d(q') - 1)$ becomes $\min(H, \rho(q') + 1)$ and the min/max exchange. $\square$

Proposition 6 is bookkeeping, but it explains the rest, since standard facts about attractor ranks yield everything below, including the bound of $\min(|Q|, H)$ synchronous iterations to reach the fixed point.

Lemma 7 (Strict progress). *At a P -owned key with $D_P(q) \geq 0$ choose any action that wins at clock $D_P(q)$; every such quiet move satisfies $D_P(q') \geq D_P(q) + 1$. At an opponent-owned key with $D_P(q) \geq 0$, every quiet move satisfies $D_P(q') \geq D_P(q) + 1$.*

Proof. Immediate from (1), since a quiet move contributes $\text{sat}(d(q') - 1)$ to the extremum that determines $D_P(q)$. $\square$

Corollary 8 (Key-positional winning strategy). *The map σ_P assigning to each winning P -key an action winning at clock $D_P(q)$ depends on the key and the masks only. From every winning (q, h) it wins, and along every play consistent with it the deadline strictly increases, so no key is visited twice.*

Proof. The invariant $h \leq D_P(q)$ holds initially and is preserved: after σ_P 's quiet move $h + 1 \leq D_P(q) + 1 \leq D_P(q')$, and after any opponent quiet move likewise by the universal clause of Lemma 7; opponent claims are unavailable at keys with $D_P \geq 0$ by (1) and $h \leq K_P(q)$; the play is finite since h increases and is bounded by $D_P \leq H - 1$; it ends at a P -won terminal or exit because those are the only actions with nonnegative value at the deadline. Strict increase of D_P along the play forbids revisiting a key. $\square$

Remark 9. Choosing an action that wins merely at the *current* clock does not give progress: a P -key with a quiet self-loop and a winning exit can loop while the exit remains available, and looping produces a repetition. The deadline-optimal choice is what makes the strategy key-acyclic.

4 The fixed-past theorem

Theorem 10 (Fixed-past threshold equivalence). *For every epoch arena, thresholds (C, H) and $(r_x, a_x)_{x \in Q}$, query (q_0, h_0, c) and target player P : P wins the explicit-history game from the query iff $h_0 \leq D_P(q_0)$ in the frozen-mask game with $A = \{c \geq a - 1\}$, $B = \{c \geq r - 1\}$. Applying this to both players, the explicit win/draw/loss value equals the frozen-mask value.*

Proof. ($\Leftarrow$) Play σ_P of Corollary 8 in the explicit game. Along any consistent play no key repeats, so at the current key q the explicit count is exactly $c(q) + 1$: a current claim or automatic draw exists in the explicit game only if it exists in the frozen-mask game, where it does not at keys with $D_P \geq 0$. A defender's *intended* claim at q concerns a quiet successor q' : if q' were on the play so far, strict progress would give $D_P(q') \leq D_P(q)$ against Lemma 7; hence q' is fresh, its explicit count after the move would be $c(q') + 1$, and an intended claim exists only if $q' \in B$, which (1) already excludes at winning defending keys. Clock claims are identical in both games. So every explicit action available to the opponent is one the frozen-mask analysis already accounted for, and the play ends in a P -win.

($\Rightarrow$) If $h_0 > D_P(q_0)$ the opponent has a counterstrategy in the frozen-mask game (a finite game). Every frozen-mask claim is a legal explicit claim, because explicit counts are at least the frozen counts plus the current occurrence; explicit automatic draws only help the opponent; moves and exits coincide. Projecting the explicit play to (q, h) , the counterstrategy remains well defined and prevents a P -win. $\square$

Corollary 11 (One mask under alternation). *If every quiet move between keys changes the owner, then after checking $q_0 \notin A$ the deadlines computed with $A = \emptyset$ agree with D_P at every key outside A ; the query needs only B .*

Proof. A winning strategy cannot enter a defending key in $A \subseteq B$ (a current claim exists there) nor a P -key in A (its predecessor is defending by alternation and has an intended claim into B). $\square$

Remark 12. The masks preserve a *value query*, not the transition system, and two pasts with equal masks diverge after one move. A certificate for one mask pair says nothing about another. The theorem is about existence of a winning strategy, not about every optimal-looking move.

5 Necessity of the hypotheses

Each of the following was checked by computing both games explicitly (`variants.py` in the released code).

Example 13 (Neutrality, rewarded repetition). One P -key with a quiet self-loop and no exit, $r = 2$, no past. If the second occurrence *wins* for the mover, the explicit value is a P -win (loop once); the frozen-mask game, which treats repetition as a draw, gives a draw.

Example 14 (Neutrality, penalised repetition). x (P -owned) $\rightarrow y$ (opponent) $\rightarrow x$ with no exits, $r = 3$, $c(x) = 1$. If creating the third occurrence *loses*, the opponent is forced to return to x and loses; the frozen-mask game with $B = \emptyset$ sees a bare cycle and returns a draw.

Example 15 (Clock monotonicity). x (P -owned, self-loop, edge to y), y (opponent) $\rightarrow z$ (P -owned) $\rightarrow$ winning exit; $H = 6$; the opponent may claim only when $h \in \{1, 2\}$. Explicitly the query is a draw: moving to y early allows a claim, waiting at x triggers the automatic repetition draw. A frozen-mask analysis with monotone clock claims reports a win. Waiting is profitable only when a larger clock can help; that is exactly what monotonicity forbids.

Example 16 (Alternation for Corollary 11). $x \rightarrow y$ both P -owned, y has a winning exit, $y \in A$. With the automatic mask the arrival at y is a draw; without it P wins. No defending turn intervenes at which an intended claim could block the move. In the enumeration of Section 6, the one-mask reduction fails on exactly the 12,288 non-alternating cases and on none of the 354,902 alternating ones.

6 Verification

An implementation written from Definitions 3–4 alone (no code shared with the companion manuscript) computes the explicit-history value by memoised backward induction over (q, h, N) , the frozen-mask value by iterating (1), and a third value by backward induction over (q, h) with frozen masks. The two-key family has, per key, 2 owners $\times$ 4 quiet-successor subsets $\times$ 8 exit subsets = 64 nonterminal configurations plus 3 terminals ($67^2 = 4,489$ arenas); the three-key family restricts exits to at most one outcome ($67^3 = 300,763$ arenas). Every arena is queried at every key, every clock and every past-count vector in $\{0, \dots, a-1\}^{|Q|}$, including count vectors unreachable by legal play.

Family	Rules (r, a, C, H)	Arenas	Root queries	Proof-scenario hits	Mismatches
two-key exhaustive	(2,3,2,3)	4,489	323,208	n/a	0
two-key exhaustive	(3,5,4,6)	4,489	1,571,150	n/a	0
random, 1–9 keys	varied	6,000	295,265	n/a	0
dense random, 3–6 keys	varied	40,000	1,257,754	n/a	0
three-key exhaustive	(2,3,2,3)	300,763	93,083,904	0 [†]	0
three-key exhaustive	(3,5,4,6)	300,763	754,152,000	30,720	0

Table 1: Explicit-history versus frozen-mask values. “Proof-scenario hits” counts winning queries in which, along the key-positional strategy, some defending key has a quiet successor lying in the supplied past one occurrence below the claim threshold and off the realised play, the case the forward direction of Theorem 10 turns on. [†]With $r = 2$ that case cannot occur.

The table totals 850 million queries with no disagreement. The implementations also let a count sit on a terminal slot, in which case a quiet move to that terminal at threshold carries an intended

claim in both games; such queries are included and never disagreed. The certificate of the deadline vector was tested separately, with a fixed seed, on 20,000 random arenas with up to 12 keys and random thresholds; the true vector passed the local fixed-point test in every one of 40,000 cases and every one of 2,486,548 single-entry mutations of it was rejected. All of this is consistency between implementations of the same definitions, not a proof, and it says nothing about whether a particular game’s rules are modelled correctly.

7 Relation to prior work

The problem is the graph-history interaction [1, 2, 3]; the classical fact that a winner can avoid repeating a position goes back to Zermelo and Kálmár [7]; attractor ranks and positional determinacy of reachability games are textbook [5, 6]; first-cycle games show that memoryless determinacy fails for general cycle objectives [8], which is why neutrality matters here; de Man’s observation that metric-optimal play avoids new repetitions while supplied history remains a separate concern is the practical antecedent [4]. Ewerhart’s Theorem 2 shows that the value of chess with optional repetition claims, made at a third occurrence or before the move that would produce one, equals the value of the finite game drawn at the third occurrence, and his appendix defines the finite updateable state, the position with the set of positions seen once, the set seen at least twice and the quiet count capped at 100, under which subgames are isomorphic [10]. Both claim forms and the count-set representation are therefore antecedents. What is new here is the fixed-query reduction, in which the seen-once set is discarded, the seen-twice and automatic sets are frozen instead of updated, the clock enters through a deadline, and exits and automatic rules are included. Hamkins announces that winning and drawing tactics in chess need no information about repetition or the fifty-move rule [9]; only the introduction of that post is public. Corollary 8 is the winner’s half of such a statement for arbitrary supplied pasts, with the past entering through the masks.

Reproducibility

The solvers, the exhaustive-run outputs, the counterexample script and the property checks are released with the companion manuscript.

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